

Calculus Skeleton Notes

Contents

September 9: Average Rates of Change	1
September 11: 1.1 - Limits	3
September 15: 1.3 - Finding Limits Analytically	7
September 16: 1.5 - Continuity	11

Average Rates of Change

Goals: Compute average rates of change, slopes of secant lines, and motivate the idea of limits

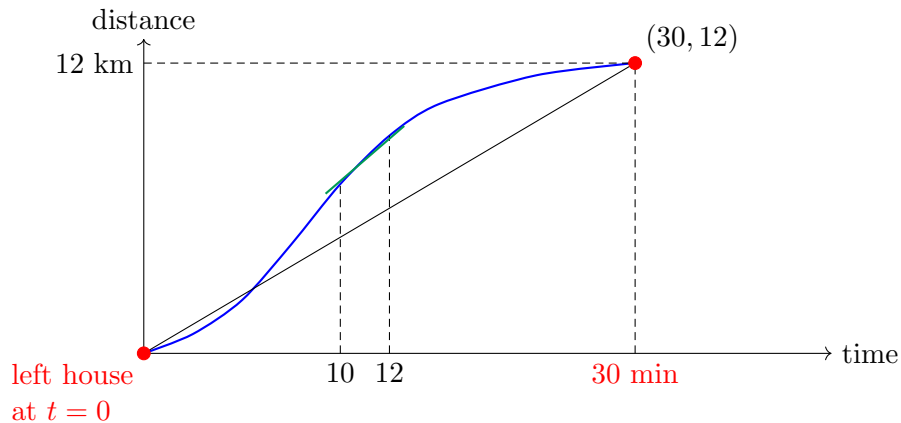
Before Class Videos: None today, starting next class

Example 1: I left my house at time $t = 0$. 30 minutes later, I was at UVic which is 12 km away.

a) What was my average speed?

$$\text{Average Speed} = \frac{\Delta \text{ distance}}{\Delta \text{ time}} = \frac{12 \text{ km}}{0.5 \text{ h}} = 24 \text{ km/h.}$$

b) Sketch a possible graph



c) How could I approximate my speed at $t = 10$ minutes?
Starting at 10 min, I go 1 km in 2 min.

$$\text{Average speed} = \frac{1 \text{ km}}{2/60 \text{ h}} = 30 \text{ km/h.}$$

How can I approximate my speed at 10 min?

Compute average velocity over $[10, 11]$, $[10, 10.1]$, $[10, 10.01]$, etc

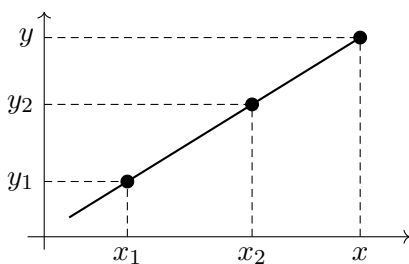
Tell LIDAR story

Future: “Need instantaneous velocity!”

Example 2: What is average rate of change of $f(x) = x^2$ on $[2, 4]$?

$$\text{Average rate of change} = \frac{f(4)-f(2)}{4-2} = \frac{16-4}{4-2} = 6$$

Example 3: Find the equation of the line through (x_1, y_1) and (x_2, y_2) .

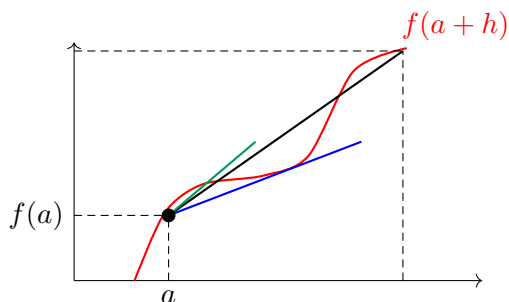


$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = m, \quad m = \frac{y - y_1}{x - x_1}.$$

$$y - y_1 = m(x - x_1) \implies y = y_1 + m(x - x_1) = mx + \underbrace{[y_1 - mx_1]}_b.$$

Example 4: Consider the graph of a function $f(x)$. How do I write an equation for the secant line through $(x_1, f(x_1))$ and $(x_2, f(x_2))$?

Answer $y = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$



Slope of secant line

$$= \frac{f(a+h) - f(a)}{(a+h) - a} = \frac{\text{rise}}{\text{run}}.$$

Big Idea: As x_2 gets closer to x_1 the slope of the secant line approaches the slope of the tangent line at x_1 . We need to investigate and define this limiting process more precisely. This DESMOS visualizes the idea: <https://www.desmos.com/calculator/oi7poyhodr>

Suggested after class problems from APEX: None today, but review the Course Outline on Brightspace

1.1 - Limits

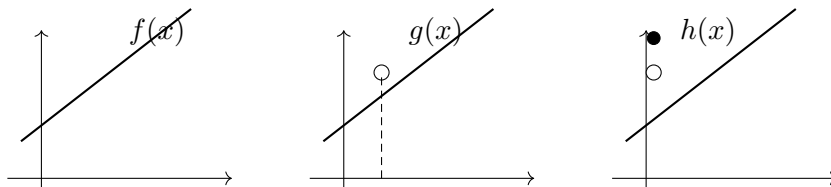
Goals: Describe limits graphically, numerically, and algebraically.

Before Class Videos: 1.1.1 and 1.1.2

Example 1: Consider these three functions:

$$f(x) = x + 1, \quad g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x + 1)(x - 1)}{x - 1}, \quad h(x) = \begin{cases} x + 1, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

How are these similar or different?



Domain: $\mathbb{R}, \quad (-\infty, 1) \cup (1, \infty), \quad \mathbb{R}.$

Same everywhere but $x = 1$, where $f(1) = 2$, $g(1)$ undefined, $h(1) = 3$.

Definition:

$\lim_{x \rightarrow a} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently close to a .

Example 2: Compute the limit as $x \rightarrow 1$ of $f(x)$, $g(x)$, and $h(x)$.

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} g(x) = \lim_{x \rightarrow 1} h(x) = 2$$

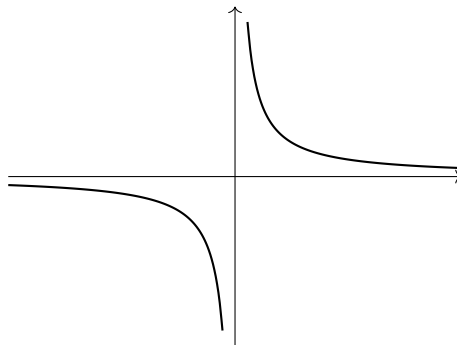
Example 3: Use a table of values to estimate $\lim_{x \rightarrow 1} (x + 1)$.

x	$f(x)$	
0.9	1.9	
0.99	1.99	
0.999	1.999	slope? closer to 2.
1.1	2.1	
1.01	2.01	
1.001	2.001	

Example 4: Use a table of values to estimate $\lim_{x \rightarrow 0} \frac{1}{x}$.

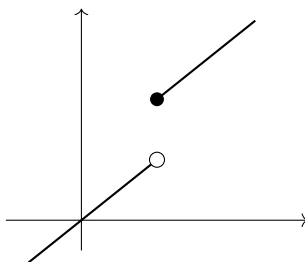
x	$f(x)$	
0.1	10	not getting close to anything!
0.01	100	
0.001	1000	

So limit does not exist!



Example 5: Investigate

$$f(x) = \begin{cases} x, & x < 1, \\ x + 1, & x \geq 1. \end{cases}$$

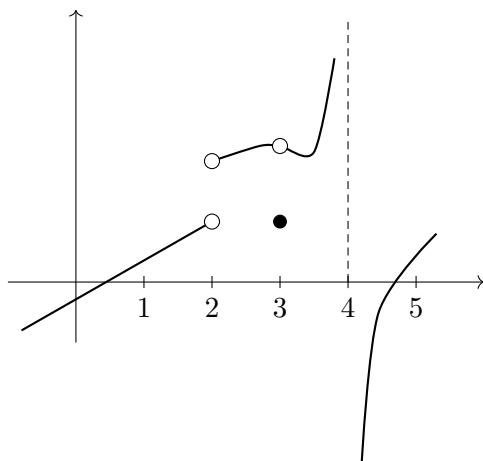


Definition:

$\lim_{x \rightarrow a^+} f(x)$ means $f(x)$ gets arbitrarily close to L for all x sufficiently close to a from the right,
i.e. $x > a$.

$$\lim_{x \rightarrow 1^+} f(x) = \quad \lim_{x \rightarrow 1^-} f(x) =$$

Example 6: Use the graph to evaluate the indicated limits.



$$\lim_{x \rightarrow 2^-} f(x) = \text{_____}, \quad \lim_{x \rightarrow 2^+} f(x) = \text{_____}, \quad \lim_{x \rightarrow 2} f(x) = \text{_____},$$

$$\lim_{x \rightarrow 3^-} f(x) = \text{_____}, \quad \lim_{x \rightarrow 3^+} f(x) = \text{_____}, \quad \lim_{x \rightarrow 3} f(x) = \text{_____},$$

$$\lim_{x \rightarrow 4^-} f(x) = \text{_____}, \quad \lim_{x \rightarrow 4^+} f(x) = \text{_____}, \quad \lim_{x \rightarrow 4} f(x) = \text{_____}.$$

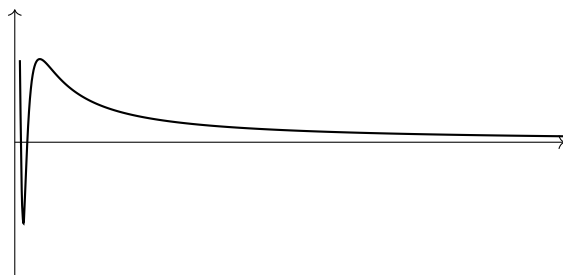
$$\lim_{x \rightarrow 2^-} f(x) = 1, \quad \lim_{x \rightarrow 2^+} f(x) = 2, \quad \lim_{x \rightarrow 2} f(x) \text{ DNE},$$

$$\lim_{x \rightarrow 3^-} f(x) = 2, \quad \lim_{x \rightarrow 3^+} f(x) = 2, \quad \lim_{x \rightarrow 3} f(x) = 2,$$

$$\lim_{x \rightarrow 4^-} f(x) \text{ DNE}, \quad \lim_{x \rightarrow 4^+} f(x) \text{ DNE}, \quad \lim_{x \rightarrow 4} f(x) \text{ DNE}.$$

Example 7: Investigate limits of

$$f(x) = \sin\left(\frac{1}{x}\right)$$



$$\lim_{x \rightarrow 0^+} f(x) \text{ DNE, as } f(x) \text{ gets close to every } y \text{ value in } [-1, 1].$$

Example 8: Graphically investigate limits of $\frac{x^2-1}{x^2-x-2}$

Suggested after class problems from APEX: Section 1.1 problems 9,11,15

1.3 - Finding Limits Analytically

Goals: Evaluate limits using limit laws, algebra, and the Squeeze Theorem.

Before Class Videos: 1.3.1–1.3.3

Definition:

If $f(x)$ gets arbitrarily close to L for all x sufficiently close to a , then

$$\lim_{x \rightarrow a} f(x) = L.$$

For many “nice” functions, evaluating a limit just means plugging in the number. We’ll investigate what functions are “nice” in section 1.5 on Continuity. For example:

Example 1:

- $\lim_{x \rightarrow 2} x^3 =$
- $\lim_{x \rightarrow \pi} \sin(x) =$
- $\lim_{x \rightarrow e} \ln(x) =$
- $\lim_{x \rightarrow 4} \sqrt{x} =$

Theorem:

If L , M , c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M,$$

then

- | | |
|----------------------------|---|
| 1. Sum Rule: | $\lim_{x \rightarrow c} [f(x) + g(x)] = L + M$ |
| 2. Difference Rule: | $\lim_{x \rightarrow c} [f(x) - g(x)] = L - M$ |
| 3. Constant Multiple Rule: | $\lim_{x \rightarrow c} [kf(x)] = kL$ |
| 4. Product Rule: | $\lim_{x \rightarrow c} [f(x)g(x)] = LM$ |
| 5. Quotient Rule: | $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$ |
| 6. Power Rule: | $\lim_{x \rightarrow c} [f(x)]^n = L^n, \quad n \text{ a positive integer}$ |
| 7. Root Rule: | $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n \text{ a positive integer}$ |

If n is even in the Root Rule, assume that $f(x) \geq 0$ on an interval containing c .

Example 2: Evaluate

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x + 2}.$$

Example 3: Come up with functions f and g such that

$$\lim_{x \rightarrow 3} f(x)g(x) \neq \left[\lim_{x \rightarrow 3} f(x) \right] \left[\lim_{x \rightarrow 3} g(x) \right].$$

Four Algebraic Tricks

Example 4:

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x - 2}.$$

Example 5:

$$\lim_{x \rightarrow 0} \frac{(x - 2)^2 - 4}{x}.$$

Example 6:

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}.$$

Example 7:

$$\lim_{x \rightarrow 2} \frac{\frac{1}{2} - \frac{1}{x}}{x - 2}.$$

Example 8: Evaluate

$$\lim_{x \rightarrow 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}}.$$

Theorem:

Squeeze Theorem. If $f(x) \leq g(x) \leq h(x)$ for all x in an interval containing a (except possibly at a), and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L,$$

then

$$\lim_{x \rightarrow a} g(x) = L.$$

Example 9: Evaluate

$$\lim_{x \rightarrow 0} x^2 \sin(1/x^2).$$

Example 10: Suppose

$$-x^2 + 4x + 1 \leq g(x) \leq -2x + 10$$

for all x . For what value of a can you determine $\lim_{x \rightarrow a} g(x)$? What is the limit?

1.5 - Continuity

Goals: Determine where a function is continuous, build continuous piecewise functions, and apply the Intermediate Value Theorem.

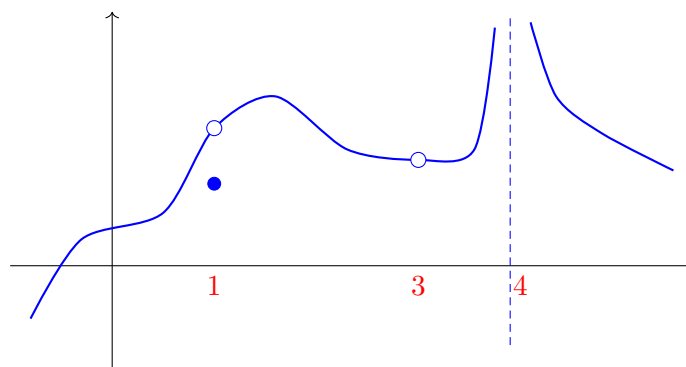
Before Class Videos: Section 1.5 videos 1 and 3

Definition:

A function f is **continuous at** c if all three of the following hold:

1. $f(c)$ exists,
2. $\lim_{x \rightarrow c} f(x)$ exists, and
3. $\lim_{x \rightarrow c} f(x) = f(c)$.

Example 1: Identify the type of discontinuity at each marked x -value.



Example 2: For what values of a is

$$f(x) = \begin{cases} x + 2, & x \leq 1, \\ ax^2, & x > 1 \end{cases}$$

continuous everywhere?

Example 3: For what values of a and b is

$$f(x) = \begin{cases} -2, & x \leq -1, \\ ax - b, & -1 < x < 1, \\ 3, & x \geq 1 \end{cases}$$

continuous everywhere?

Combining Continuous Functions

Theorem:

If f and g are continuous at c , then $f + g$, $f - g$, constant multiples of f , and fg are continuous at c . The quotient f/g is also continuous at c provided $g(c) \neq 0$.

Let's justify the sum rule from the limit laws:

Composition of Functions

Example 4: Compute the domain and range of $\sin \sqrt{x}$

Theorem:

Suppose f is continuous at a and g is continuous at $f(a)$. Then the composition $g \circ f$ is continuous at a , and

$$\lim_{x \rightarrow a} g(f(x)) = g(f(a)).$$

Example 5: On what intervals is $\sin\left(\frac{x-2}{x^2+5x+7}\right)$ continuous?

Example 6: Evaluate

$$\lim_{x \rightarrow 1} \frac{\sin(\pi x/6)e^{x^3} + x}{x^2 + 2}.$$

Theorem:

Intermediate Value Theorem. If f is continuous on $[a, b]$ and y_0 is between $f(a)$ and $f(b)$, then there is some $c \in [a, b]$ such that

$$f(c) = y_0.$$

Sketch:

Example 7: Sketch examples where the assumptions of the Intermediate Value Theorem are not satisfied.

Example 8: Show that

$$x^3 - 3x - 1 = 0$$

has a solution in $[0, 2]$.

Example 9: Use the intermediate value theorem to show that at some time a baby's height equalled their child's height.

Suggested after class problems from APEX: Section 1.5 problems 21,24,35